Story Setup: Combined Waves in Real Flows

AERO-DYNAMICS II

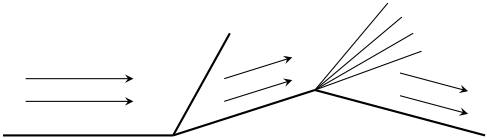
High-Speed Compressible Flows

Why Combined Waves Are the Norm

- Supersonic flow cannot adjust smoothly to geometry changes.
- Every turn forces a **local wave**:
  - concave turn $\rightarrow$ compression (oblique shock),
  - convex turn $\rightarrow$ expansion fan.
- Real geometries have *multiple* turns, so waves appear in **sequence**.
- The downstream state depends on the **order** of those waves.

Two-Turn Example: Compression Then Expansion

- First turn: flow turns *toward* itself $\rightarrow$ oblique shock.
- Second turn: flow turns *away* $\rightarrow$ expansion fan.
- The expansion acts on the **post-shock** state, not the freestream.
- This is the root of **order sensitivity**.



Quick Review: What Waves Do

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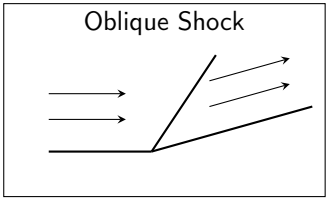
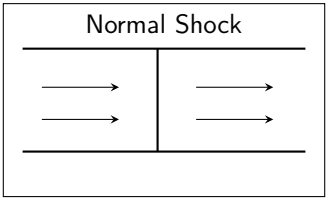
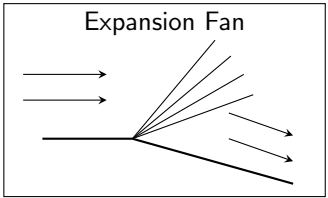
Wave Types and Core Effects

- **Oblique shock:** compression with flow turning toward a surface.
- **Normal shock:** strong compression; downstream becomes subsonic.
- **Expansion fan:** isentropic turning away from a surface.

Wave Type	Turns Flow	Raises p	Raises M	Entropy Increase
Oblique Shock	Yes	Yes	No	Yes
Normal Shock	No	Yes	No	Yes
Expansion Fan	Yes	No	Yes	No

Canonical Wave Sketches